Interpretable Machine and Deep Learning Framework for Tea Leaf Disease Detection

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Abstract

Tea is one of the most significant cash crops in Bangladesh, yet its production is severely threatened by leaf diseases such as Tea Leaf Blight, Tea Red Leaf Spot, and Tea Red Scab. Early and accurate disease detection is critical for sustaining crop yield, quality, and farmer livelihoods. This study presents a comparative framework that integrates both machine learning (ML) and deep learning (DL) techniques, enhanced with Explainable Artificial Intelligence (XAI), to detect tea leaf diseases with high accuracy and interpretability. A dataset of 253 high-resolution images collected from Bangladeshi tea gardens was used, comprising healthy leaves and three diseased classes. Multiple ML classifiers (KNN, SVM, Decision Tree, Random Forest) and DL models (CNN, VGG16, VGG19, ResNet34, MobileNetV2, AlexNet, EfficientNetB0) were evaluated. Among these, EfficientNetB0 achieved the highest accuracy of 96%, outperforming other models in precision, recall, and F1-score. To enhance trust and usability, XAI methods—LIME and Grad-CAM—were employed to highlight critical image regions influencing predictions, making the system transparent for both experts and farmers. A user-friendly web application was developed using Streamlit, enabling real-time disease detection by simply uploading leaf images. The findings not only demonstrate the potential of AI-driven models for precise tea disease classification but also emphasize their role in promoting sustainable agriculture through reduced pesticide misuse and improved decision-making. This research bridges the gap between advanced AI models and practical agricultural applications, offering an interpretable and accessible solution for tea disease management in Bangladesh.

Keywords:
Tea Leaf Disease Detection, Real-Time Monitoring, Deep Learning, Machine Learning, Explainable AI, Web Application

1. Introduction

Tea is one of the prominent cash crops in the world which has been contributing to the economic growth of many countries. Bangladesh has been contributing to world tea production since 1984 and accounts for 1.63% of total production [1]. However, the growth of tea gets hampered when the plants are affected by various diseases such as Algal leaf spot, Gray blight, White spot, Brown blight, Red scab, Bud blight, and Grey blight etc [2]. Around 10-15% of tea crops in Bangladesh get affected by these diseases every year [3]. Early detection of tea leaf disease is an essential factor to solve this problem which is often time consuming and requires expert knowledge if done manually. In recent years, machine learning and deep learning have reached the zenith of the technological innovation while detecting and classifying plant diseases. The convolutional neural network (CNN) associated with Segment Anything Model [4], ADAM optimizer [5], [6], MobileNetV2 [7], enhanced MobileNetV3 model [8] etc were able to detect the most common diseases of tea leaves with an accuracy of 95.06%, 84%, 99%, 94.55% and 95.88% respectively. Some object detection models such as- YOLOv7 [3], BRA-YOLOv7 [9], YOLOv8 [10], an improved RetinaNet target detection and identification network-AX-RetinaNet [11] achieved 97.3%, 94.9%, 98% and 93.83% accuracy correspondingly in order to distinguish multi-class disease from healthy leaves. Nevertheless, the models used in these studies functioned as a "black box" shedding very little light on decision making process. Farmers frequently have to abide by laws created by the system without knowing how to interpret or validate them. This demonstrates the pressing need for reliable, site-specific methods to identify diseases linked to tea. Explainable artificial intelligence plays a vital role here to provide interoperability and transparency to the model accuracy [12]. Addressing the challenge, the authors of [13] found out that the GradCAM is the most suitable method to detect several plant leaves diseases because it can assign higher weight to the location of lesion. However, in case of tea leaf disease detection there is only one study found where XAI technique was implemented [14]. Here, a modified Xception-based CNN model was developed acquiring 99.71% accuracy and LIME and GradCAM explained the decision-making process. To add significant value to the research of tea leaves disease detection, a mobile application is developed [15] enabling farmers to identify disease at early stage using conventional CNN with more than 92% average accuracy. To carry out our research, the dataset is collected from tea gardens of different locations of Bangladesh. The goals of the study are: To develop a web based efficient system using ML and DL models that achieves high-accuracy disease detection of tea leaves To provide a clear, understandable explanations for end-users with the help of XAI. In order to extract features automatically, the CNN is implemented where VGG16, VGG19, AlexNet, ResNet34, MobileNetV2, and EfficientNetB0—are employed. In addition to that, K-Nearest Neighbors (KNN), Support Vector Machine

(SVM), Decision Tree (DT), and Random Forest etc. ML classifiers are used to extract features manually. To make the classifier model more trustworthy and transparent, two explainable AI model i.e. LIME and GradCAM are implemented. Finally, a web interface is designed where a farmer can easily identify the type of disease of a tea leaf by uploading the picture of that leaf only. This study emphasizes on the instant need for strong, onerous however site-specific solutions for tea-associated diseases testing. This study attempts to fill the current gaps with a fusion of deep learning-based XAI as LIME and GRAD-CAM with CNN. The objective is to develop a system that functions as an efficient tool for high accuracy detection of the diseases with clear description to the end-user so that the excess use of better chemicals can be minimized and confidence can be built in the industry on the Bangladeshi tea and sustainable agricultural development.

1.1. Objective

The main objective of this research work are:

1. **Develop an accurate tea leaf disease detection system** by integrating machine learning (ML) and deep learning (DL) models for healthy and diseased classes (Tea Leaf Blight, Tea Red Leaf Spot, Tea Crimson Scab).
2. **Enhance model interpretability and trust** using Explainable AI (XAI) techniques such as LIME and Grad-CAM to provide clear, instance-level explanations for predictions.
3. **Create a practical, user-friendly application** that enables farmers and agricultural experts to perform real-time disease detection and make informed decisions, supporting sustainable tea cultivation in Bangladesh.

2. Literature Review

Both Explainable AI and Machine Learning in tea disease diagnosis have made a major leap to provide new tools for agriculture advancement. To elucidate these issues, this review summarizes 20 representative studies, describing their methods, results, and constraints. Ahmed et al. [1] came up with the precision in diagnosing tea complaints, the ninjas of image segmentation: YOLOv7 based Tea Disease Detection and Identification (YOLO-T) and published in “YOLOv7 based Tea Disease Detection and Identification (YOLO-T)” that adds the delicacy, hitting all classes as they use multiple models to get this including YOLOv7, CNN, Deep CNN, DNN, AX-RetinaNet, YOLOv5 and their findings indicate an enhancement in detection accuracy, and the efficacy of this approach assists in identifying the diseases. On the downside, the study also identifies some of the significant issues in the area, including long runtime, a lack of available data, image quality, poor implantation, IoT deployment issues, non-machine learning techniques, and inadequate testing. These limitations illustrate the necessity of yet further optimization and usability; work to generate that usability and efficiency in such systems. This study thus provides significant light for identifying agricultural diseaseby the use of ML. Rahman et al. [2] further branches to several other approaches presented in a systematic overview “Machine Learning-Based Tea Leaf Disease Detection: A Comprehensive Review”. It compares the performance of different machine learning algorithms, such as Vision Transformer models (ICVT, GreenViT, PlantXViT), DenseNet, ResNet-50V2, YOLOv5, YOLOv7, CNN, MobileNetv2, and Lesion-Aware Visual Transformer, for tea leaf disease detection.Existing approaches - as the authors claim - face multiple challenges like poor generalization across different datasets, limitations of small and imbalanced datasets, environmental changes affecting classification performance of ML models, and lack of real-time actionable insights. The authors highlight the importance of well-performing models that can generalize to different conditions and provide interpretable outputs to assist agricultural decision making. These insights emphasise the need to address the existing gaps in current research to develop the more effective and translatable solutions for tea illness identification. Rahman et al. Nekooeih et al. [4] suggest the study on tea leaf disease and insect identification based on improved MobileNetV3 in Frontiers In Plant Science (2024). The authors had built a system that using the Enhanced MobileNetV3 Model was used to identify tea bugs and insect pests. The paper indicates that the limited focus on adult-stage pests, challenges in generalizing the model for other geographic regions, potential low false-positive ratios due to visual similarities between many diseases and insect damage made it difficult to compare apples and oranges, and a relatively limited scope for translation to other crop pests and diseases. This implies the necessity of conducting more studies to broaden the model’s generalizability and relevance in various agricultural industries. Nekooeih et al. [4] suggest the title “YOLOv8-RCAA: A Lightweight and High-Performance Network for Tea Leaf Disease Detection”. This method is implemented by the authors using optimized YOLOv8 architecture to identify good tea bugs. The proposed method comprises of the introduction of Residual Channel Attention Aggregation (RCAA) that enhances the extraction capacity without making the architecture heavy. Recognizing that this was the only test of the models is an important cornerstone of the models themselves. The system optimized for speed and performance can be used in UAV (drone) and IoT (Internet of Things) system integrations instantly. The authors emphasize a variety of the trained datasets to perform better in real-world scenarios, and become more versatile in agriculture. Nahiduzzaman et al. [5] (Automated Mulberry Leaf Disease Classification — Explainable Deep Learning Model) propose an explainable deep learning model. The authors utilized deep learning techniques to develop a method to identify diverse diseases affecting mulberry leaves important for the production of silkworms. The model is built on CNN models and literature proposes general descriptive intelligence (XAI) to provide a common take away for the decision making process. This definition enhances reliability and validity by enabling the users to identify the area of the mulberry leaf that affects the classification model. Thereafter, the precision, accuracy, recall, and F1 scores were calculated to evaluate the performance of the system, depicting the classification of different classes of leaf diseases by the system. Rahman et al. [6] “Automated

Detection of Selected Tea Leaf Diseases in Bangladesh with-Convolutional Neural Network (2024)", designed a system for detection of disease in Bangladesh. The approach employed a dataset which was limited in scope and was neither utilized with contemporary techniques like transfer learning with state of the art CNN architectures. Among those diseases, identification depended in part on microscopic confirmation, highlighting a need for flexibility at the field level. Despite the potential capabilities displayed by the CNN, the system acknowledged limitations of the figures and complexity in the data as well as CNN architecture, indicating prospects for improvement in future work. Ammar Oad et al. [7] developed a method for the disease classification over plant leaf using ensemble learning and eXplainable AI (XAI) under the paper title "Plant Leaf Disease Detection Using Ensemble Learning and Explainable AI". In order to classify plant pathogens more accurately, the authors developed a framework that allows multiple machine learning models to compete in a breeding programme to perform better on the classification task. They will adopt XAI towards making the system explainable. Cluster learning not only helps in preparing category-based data for disease classification, but also helps in making better generalised models that contain more accurate classification in the areas of interest; and explainable artificial intelligence (XAI) adds an interpretable forward model of the decision process and makes it applicable to reality in practice in the agriculture domain. Rahman et al. [8] nda system of title: "Method for the Classification of Tea Diseases by Weighted Sampling and Hierarchical Classification Learning." The authors used a weighted approach to handle the data imbalance problem, and then developed a hierarchical classification learning model for tea leaf disease classification. The paper tackles the common problems that occur in agricultural datasets such as overfitting on the training data, balance of data (hence returning incorrect predictions), and a small set of training data that lacks diversity. This technique applies weighted sampling to ensure that each disease class is represented in a proportionate manner in the resulting training dataset for the model. The models are further enhanced to allow hierarchical classification learning to classify multiple diseases, thus increasing the classification power. Experimental Results on a high informative imbalanced dataset indicate a significant improvement on classification accuracy when using the new measures. The study significance is that, for the implementation of robust and productive models for the applications in real-world settings in agriculture, the constraints of the dataset should be acknowledged. Chen et al. [9] identify a system entitled "Integrated Learning-Based Pest and Disease Detection Method for Tea Leaves". The authors proposed a systematic methodology that encompasses various machine learning techniques employed for the detection of pest and disease. The paper points out the major problems such as how to deal with historical background in images, precisely identifying small targets, deploying models efficiently, and tuning parameters for maximization. Despite these limitations, our proposed work shows that integrating machine learning can assist in accurate and scalable detection. We listed this project at MDPI, in 2023. Rahman et al. In 2024[10], "YOLO-Tea: A Deep Learning Based Tea Disease Detection System,"[10] proposed a system "YOLO-Tea: A Deep Learning Based Tea Disease Detection System". To overcome problems with detecting small targets, complicated lighting conditions, and environmental restrictions in tea gardens, The authors had created a methodology which combines the YOLOv5s model with ACmix, CBAM, RFB, and GCNet modules. Lastly, this system also solves the problem where there is no mechanism for the accurate and actionable delivery of pesticide. Their findings provide a strong approach to enable early disease detection under difficult conditions, thus helping improve tea crop management. Kumar et al. [11] proposed a system for "Grey Blight Disease Detection Using DCNN" It employs a DNN (Deep Convolutional Neural Network) to identify Grey Blight disease considering the tea leaves. The authors note limitations like a small dataset, focusing on one disease (Grey Blight), and hurdles regarding model complexity and deployment. This work also highlights the necessity of a more extensive dataset and more straightforward deployment strategies to improve practical applications.

Zhao et al. [12] for "Tea Leaf Blight Detection Using DDMA-YOLO with UAV Remote Sensing". This method fuses images obtained from UAV to DDMA-YOLO to Detect Tea Leaf Blight. On the other hand, few drawbacks have to be mentioned, most prominently dependence on certain UAV images and environments, insignificant performance gain with the introduction of an attention module, and a single disease disease this work was focused on. Issues like possible overfitting and poor generalization for the models are also mentioned, while it is asked for adaptable models and more datasets. Ahmed et al. [13], present a study on "Food Security and Dietary Diversity of Tea Workers in Two Gardens of the Greater Sylhet District of Bangladesh," where dietary diversity and food security were assessed for tea workers from two tea estates in the greater Sylhet district of Bangladesh. This research investigates the socio-economic challenges for tea workers and the link between food diversification and food security. As with previous works of theirs this book highlights areas in terms of worker welfare and nutrition where improvement is needed, along with enlightening analysis in terms of the socio-economic issues / challenges faced by the tea industry. Tiwari et al. [14] multi-class plant disease detection using dense cnn's The authors utilize a Dense Convolutional Neural Network (DCNN) model to identify and categorize different diseases of plants from mere images of leaves. They conclude with positive aspects of the particular system in improving early detection and enhancing agricultural practices but point out the obstacles involved like the sheer size of datasets and the eventual increase in model complexity undermining practical applications. Findings highlight scalability and usability as key to optimizing applied implications for agriculture. Zhang et al. [15], introduce a system called "Predicting Plant Disease Outbreaks Using Machine Learning Models", which utilizes environmental conditions for predicting disease outbreak risks. Owing to the greater utility of this method, potential applications such as enhancing tea with favorable production traits can help realize precision agriculture. Li et al. [16] is a proposed system for Plant Disease Classification Using CNNs and Machine Learning. This ap-

proach integrates the convolutional neural networks with machine learning algorithms to identify multi-plant disease classification. While the study highlights the promise of the system for the detection of disease very early and intervention in farmers, problems including limited datasets and complexities associated with deploying it were also apparent. The authors also push for more research on increasing dataset diversity and the development of ways to facilitate deployment for practical use. Ahmed et al. [17] combine transfer learning to propose "A Small Target Tea Leaf Disease Detection Model." In our model, we apply transfer learning methods to achieve better detection of more subtle indicators of disease in tea leaves. The authors mention limitations like model overfitting from small datasets and difficulties with high precision for small target diseases. They want more stable training methods and diverse datasets to do better. Zhao et al. [18] Many have tried developing methodologies for leaf disease detection, while Demiriz and Ekenel et al. Models used for detection include CNN, VGG, ResNet, SVM, Random Forest, MTS VM, etc. Some of the pioneers who sit on their supply chain leadership team transition connected systems technologies to standard production systems and processes. The study highlights the importance of steadily improving dataset quality and model interpretability for more advanced deployment in real-world agricultural applications. The year of publication was 2023. Kumar et al. UNFOLDLY [19] this system "DLMC-Net: Deeper Lightweight Multi-Class Classification Model for Plant Leaf Disease Detection" is introduced as a convolutional neural network with multiclass classification architecture for numerous plant leaf disease detection. The authors acknowledge a number of limitations, such as a narrow range of disease coverage, dependence on data augmentation, and assessment on small sets. Moreover, the study does not benchmark against a wider range of models and has limited explainability. CNN, VGG, ResNet, SVM, Random Forest, and MTSV[Reference from Previous Work]. To continue, these limitations could be addressed to improve the usability and robustness of the model. Heng et al. [20] introduce a system entitled "A New AI-Based Approach for Automatic Identification of Tea Leaf Disease Using Deep Neural Network Based on Hybrid Pooling." In this study, the authors propose a methodology to diagnose diseases in leaves of tea plants automatically using a hybrid pooling deep neural network. The study highlights key barriers, including enhancing detection accuracy and addressing differences in symptomatology across various disease phenotypes. It is reported how efficient deep learning approaches are in improving disease detection performance. We have done unique work on this project after a solid study of the papers as mentioned above. No papers have been found that utilized machine learning algorithms and deep learning algorithms simultaneously. But at the same time we used four ML algorithms and seven DL algorithms for best results. We thoroughly compared performance to select the best performing algorithm according to the accuracy, both on ML and DL models and also within each grouping. Another unique aspect of our dataset is its origin, collected from Bangladesh tea gardens. The Explainable AI (XAI) applied for transparency and interpretability We have already utilized LIME for ML models and Grad-CAM (Gradient-weighted Class Activation Mapping) for DL models. Finally, we have built an easy-to-use web application using Streamlit. Used for this web app Upload tea leaf images and the app shows users the prediction.
Table 1 provides a summary analysis of related research works done in the year 2024.

3. Methodology

3.1. System Design

The proposed methodology of our system is given in figure 1. The figure illustrates how they use ML and DL techniques to detect tea leaf diseases. Here is a detailed description of each of these steps in the process:

- **Dataset Collection:** A dataset of tea leaf images is collected, including both healthy and diseased leaves across multiple categories.
- **Preprocessing:** The images undergo several preprocessing steps:
  - **Image Resizing:** Standardizes all images to a uniform size.
  - **Image Augmentation:** Generates variations of images (e.g., rotation, flipping) to expand the dataset and improve model generalization.
  - **Image Splitting:** Divides images into smaller, manageable sections for training.
- **Training Phase:** Two approaches are used:
  - **Deep Learning Networks:** Models such as AlexNet, VGG16, VGG19, ResNet34, MobileNetV2, and EfficientNetB0 automatically extract features and classify images.
  - **Manual Feature Extraction with ML Classifiers:** Features are manually extracted and fed into machine learning classifiers, including Random Forest, KNN, SVM, and Decision Trees (DT).
- **Testing Phase:** The trained classifiers are evaluated on a separate test set to assess their disease prediction performance.
- **Explainable AI (XAI):** XAI techniques are applied to provide interpretable explanations of model predictions, highlighting the features influencing each classification.

3.2. Data Collection

For our research, we gather the dataset from tea gardens in different districts of Bangladesh; the dataset contains tea leaf diseases that affect local crops. It is the images of tea leaves of four classes: Healthy, Tea Red Leaf Spot, Tea Leaf Blight, Tea Red Scab. The dataset consists of 253 high-quality images, 73 healthy leaves images and 60 images for each class of the disease. This dataset is uniquely compiled to address the objective of identifying and classifying tea leaf ailments using machine

Table 1: Summary of Tea Disease Detection Studies

Reference	Methodologies	Contributions	Limitations
Ahmed et al. [1]	YOLOv7-based Tea Disease Detection (YOLO-T), CNN, Deep CNN, DNN, AX-RetinaNet, YOLOv5	Enhanced detection accuracy using multiple models for tea disease identification	Long runtime, limited data, poor image quality, IoT deployment issues, inadequate testing
Rahman et al. [2]	Vision Transformer (ICVT, GreenViT, PlantXViT), DenseNet, ResNet-50V2, YOLOv5, YOLOv7, CNN, MobileNetv2, Lesion-Aware Visual Transformer	Comprehensive review of ML algorithms, emphasizing interpretable outputs and generalization	Poor generalization, small/imbalanced datasets, environmental variability, lack of real-time insights
Nekooeih et al. [4]	Enhanced MobileNetV3 for tea leaf disease and insect identification	Improved pest detection with focus on adult-stage pests	Limited focus on adult-stage pests, poor generalization across regions, low false-positive ratios
Nekooeih et al. [4]	YOLOv8-RCAA (Residual Channel Attention Aggregation)	Lightweight, high-performance network optimized for UAV/IoT integration	Limited dataset diversity, needs broader real-world testing
Rahman et al. [6]	CNN for tea leaf disease detection in Bangladesh	Effective detection using localized dataset	Limited dataset scope, reliance on microscopic confirmation, complexity in CNN architecture
Rahman et al. [10]	YOLO-Tea (YOLOv5s with ACmix, CBAM, RFB, GCNet)	Robust detection under complex conditions, improved early disease detection	Challenges with small targets, complex lighting, and pesticide delivery mechanisms
This Study	Four ML algorithms (SVM, Random Forest, etc.) and seven DL algorithms (CNN, ResNet, etc.),	Combined ML/DL approach, Bangladesh-specific dataset, transparent predictions via XAI,	....

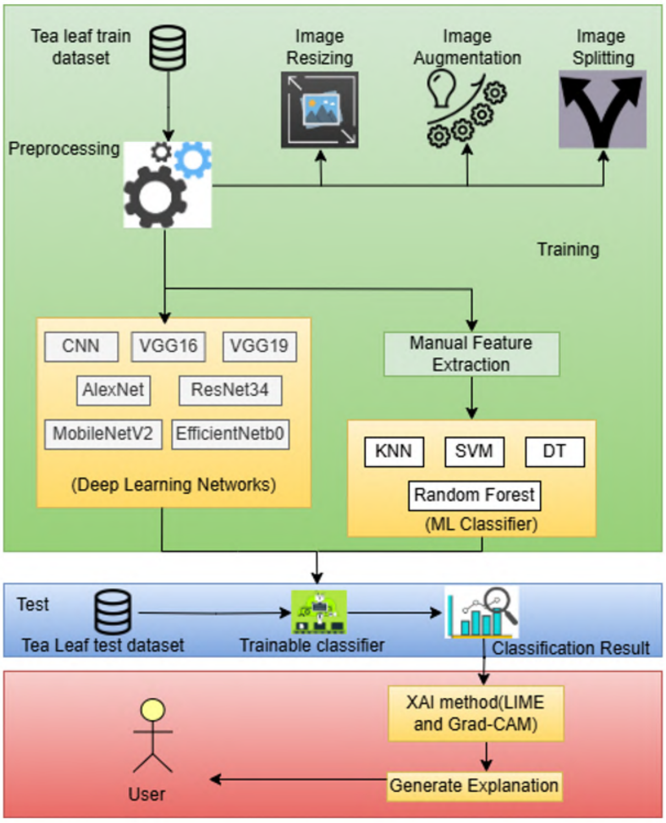


Figure 1: Proposed Methodology of Tea Leaf Disease Detection

learning techniques. The dataset allows CNN and XAI techniques like LIME and GRAD-CAM, so that we can analyze disease and understand predictions better. This tailored dataset, specific to the region as well as the disease, is crucial in building a strong and interpretable tool specific to the Bangladeshi tea industry.

3.2.1. *Description of Dataset*

As we discover tea disease in Bangladesh, the data set has different features in which we have to identify diverse disorders like Red Spot, Tea Leaf Blight, Tea Red Scab. Such attributes play an essential role in accurately classifying the health status of tea leaf leaves, and recovery through early diagnosis and management of ailments.
A detailed description of the dataset is provided in Table 2. Data source: [https://data.mendeley.com/datasets/tt2smzrzrs/1]

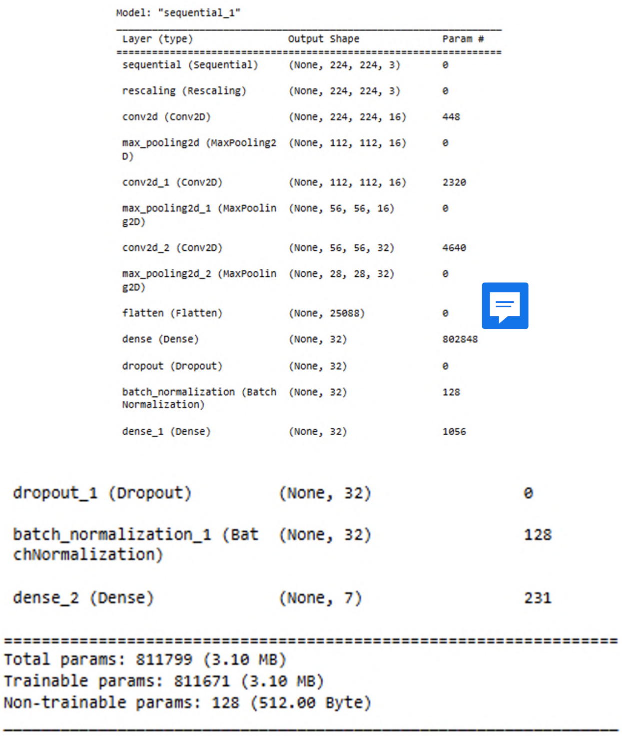
3.3. *Analyzing Data*

3.3.1. *Pre-processing of Data*

Data Preprocessing is an important step in preparing raw states for analysis or modeling. These techniques help us to enhance the performance and scalability of the model. More specifically, we use a sequential model to define the flow of layers (rescaling to normalize input data, Conv2D to extract features, MaxPooling2D to reduce dimensionality, etc.) in a neural network. We use Flatten to flatten the output from 2D to 1D, and then we use Dense layers to make predictions. So we

Table 2: Dataset Description

	Healthy	Tea Leaf Blight	Tea Red Spot	Tea Red Scab
Question	• Normal leaf appearance • vibrant color • healthy growth	• Water-soaked lesions • brown or black spots • wilting • yellow margin	• Red to purple spots with yellow halo • premature leaf droptions	• Raised reddish-brown lesions • scabby texture • distorted growth
Volume of the Dataset/ Number of image	Image Number = 73	Image Number = 60	Image Number = 60	Image Number = 60



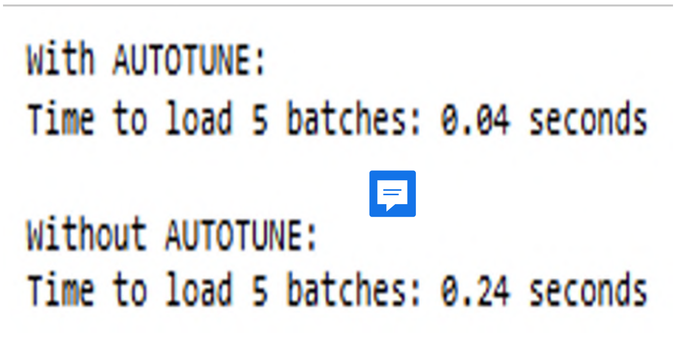
```
Model: "sequential_1"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 sequential (Sequential)     (None, 224, 224, 3)       0

 rescaling (Rescaling)       (None, 224, 224, 3)       0

 conv2d (Conv2D)             (None, 224, 224, 16)      448

 max_pooling2d (MaxPooling2  (None, 112, 112, 16)      0
 D)

 conv2d_1 (Conv2D)           (None, 112, 112, 16)      2320

 max_pooling2d_1 (MaxPoolin  (None, 56, 56, 16)        0
 g2D)

 conv2d_2 (Conv2D)           (None, 56, 56, 32)        4640

 max_pooling2d_2 (MaxPoolin  (None, 28, 28, 32)        0
 g2D)

 flatten (Flatten)           (None, 25088)             0

 dense (Dense)               (None, 32)                802848

 dropout (Dropout)           (None, 32)                0

 batch_normalization (Batch  (None, 32)                128
 Normalization)

 dense_1 (Dense)             (None, 32)                1056

 dropout_1 (Dropout)         (None, 32)                0

 batch_normalization_1 (Bat  (None, 32)                128
 chNormalization)

 dense_2 (Dense)             (None, 7)                 231

=================================================================
Total params: 811799 (3.10 MB)
Trainable params: 811671 (3.10 MB)
Non-trainable params: 128 (512.00 Byte)
_________________________________________________________________
```

Figure 2: Preprocessing of data without autotuning

use dropout to not overfit and Batch Normalization to stabilize our training. All these features allow the model to process the data adequately and find patterns that are most important for achieving high scores. All three datasets with no missing values in all columns.
Figure 2 and Figure 3 illutrates the preprocessing method of data before and after auto-tuning.

3.3.2. *Data Visualization*

Data visualization is one of the significant steps to the data analysis process that allows us to see patterns, trends, and insights that might be less visible in raw data. This research deals



```
With AUTOTUNE:
Time to load 5 batches: 0.04 seconds

Without AUTOTUNE:
Time to load 5 batches: 0.24 seconds
```

Figure 3: Preprocessing of data after autotuning

with data visualization to identify the features that contribute to (Healthy, Tea Leaf Blight, Tea Red Spot, Tea Red Scab. Thus directing further processes in the machine-learning pipeline.
For **Dataset Sample:**
Data visualization our dataset taken from various tea leaf gardens of our country. Figure 4 visualizes some of the data for our research.

3.4. *Algorithms Used*

3.4.1. *Machine Learning Algorithms*

Here, the various predictive machine learning models that are looked into are KNN, SVM, DT, Random Forest Classifier, among others. Based on the chart for classification, we further examined the accuracy, precision, recall, f1-score and support of the model. For each of these models, we then showed the confusion matrix and model assessment, together with their respective classification report-precision, recall, f1-score, support.

3.4.2. *Explainable AI*

We apply Explainable AI (XAI) techniques—LIME and Grad-CAM—to interpret model predictions for tea leaf disease detection. LIME, a model-agnostic method, perturbs input data to highlight feature contributions (e.g., color, shape, texture) for each instance. Grad-CAM, designed for CNNs, generates

Figure 4: Visualisation of the data collected from various tea gardens

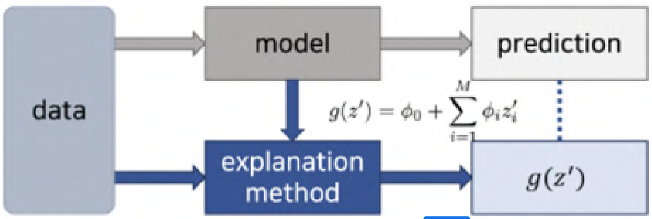


Figure 5: Concept of XAI

heatmaps that localize image regions (lesions, stains, discolorations) most influential to predictions. Together, they provide complementary, transparent, and instance-specific insights, ensuring model decisions are interpretable and useful for stakeholders in the tea sector.

3.5. Model Evaluation

We present the confusion matrix, model evaluation, classification report, and sample image predictions for the tea leaf disease dataset. The confusion matrix assesses the model's ability to distinguish between diseases like Tea Leaf Blight, Tea Red Leaf Spot, and Tea Crimson Scab. Key metrics—precision, recall, F1-score, and support—are summarized with class-wise and overall averages, while sample images show actual versus predicted classifications.

4. RESULT AND DISCUSSION

4.1. Machine Learning Algorithms

We utilized KNN, SVM, RF, DT algorithms. Their evaluation metrices are described in the below sections.
Table 4 illustrates the comparative analysis of ML algorithms performances. The overall comparison of accuracy of the Machine Learning Algorithms implemented are provided in Figure 23.

```
Confusion Matrix:
 [[13  0  0  0]
 [ 1  3  4  0]
 [ 1  0 13  0]
 [ 0  0  1 15]]

Classification Report:
                    precision    recall  f1-score   support

           healthy       0.87      1.00      0.93        13
    Tea leaf blight      1.00      0.38      0.55         8
 Tea red leaf spot       0.72      0.93      0.81        14
      Tea red scab       1.00      0.94      0.97        16

          accuracy                           0.86        51
         macro avg       0.90      0.81      0.81        51
      weighted avg       0.89      0.86      0.85        51


Accuracy: 0.8627450980392157
```

Figure 6: confusion matrix, classification report, accuracy using KNN

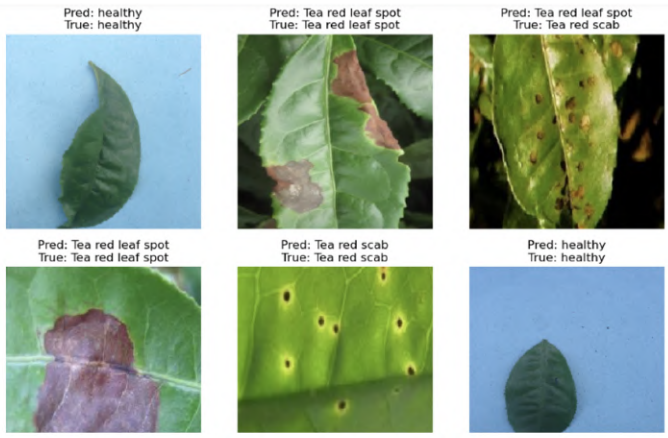


Figure 7: Prediction Result Using KNN

```
Confusion Matrix:
 [[12  0  0  0]
 [ 1  6  0  0]
 [ 0  2 11  0]
 [ 0  0  0 19]]

Classification Report:
                    precision    recall  f1-score   support

   Tea leaf blight       0.92      1.00      0.96        12
 Tea red leaf spot       0.75      0.86      0.80         7
      Tea red scab       1.00      0.85      0.92        13
           healthy       1.00      1.00      1.00        19

          accuracy                           0.94        51
         macro avg       0.92      0.93      0.92        51
      weighted avg       0.95      0.94      0.94        51


Accuracy: 0.9411764705882353
```

Figure 8: confusion matrix, classification report, accuracy using SVM

Table 3: Evaluation Metrics Comparisons of ML algorithms

Machine Learning Algorithms	Dataset	Accuracy	Precision	Recall	F1-Score	Support
K-Nearest Neighbor (KNN)	Healthy	0.8627	0.87	1.00	0.93	13
	Tea Leaf Blight	0.8627	1.00	0.38	0.55	8
	Tea Red Leaf Spot	0.8627	0.72	0.93	0.81	14
	Tea Red Scab	0.8627	1.00	0.94	0.97	16
Support Vector Machine (SVM)	Healthy	0.9411	1.00	1.00	1.00	19
	Tea Leaf Blight	0.9411	0.92	1.00	0.96	12
	Tea Red Leaf Spot	0.9411	0.75	0.86	0.80	7
	Tea Red Scab	0.9411	1.00	0.85	0.92	13
Decision Tree Classifier (DT)	Healthy	0.8627	1.00	0.92	0.96	13
	Tea Leaf Blight	0.8627	0.83	0.62	0.71	8
	Tea Red Leaf Spot	0.8627	0.80	0.86	0.83	14
	Tea Red Scab	0.8627	0.83	0.94	0.88	16
Random Forest (RF)	Healthy	0.9411	1.00	1.00	1.00	13
	Tea Leaf Blight	0.9411	0.88	0.88	0.88	8
	Tea Red Leaf Spot	0.9411	0.87	.0.93	0.90	0.14
	Tea Red Scab	0.9411	1.00	0.94	0.97	16

Figure 9: Prediction Result Using SVM

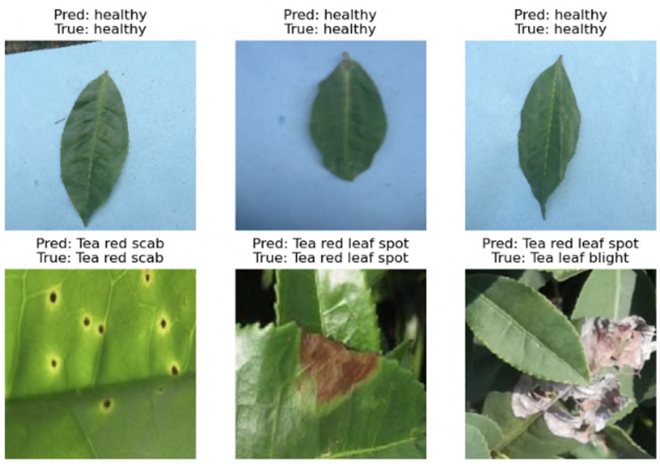


Figure 11: Prediction Result Using DT

```
Confusion Matrix:
 [[12  0  1  0]
 [ 0  5  2  1]
 [ 0  0 12  2]
 [ 0  1  0 15]]

Classification Report:
                   precision    recall  f1-score   support

          healthy       1.00      0.92      0.96        13
  Tea leaf blight       0.83      0.62      0.71         8
Tea red leaf spot       0.80      0.86      0.83        14
     Tea red scab       0.83      0.94      0.88        16

         accuracy                           0.86        51
        macro avg       0.87      0.84      0.85        51
     weighted avg       0.87      0.86      0.86        51


Accuracy: 0.8627450980392157
```

Figure 10: confusion matrix, classification report, accuracy using DT

```
Confusion Matrix:
 [[13  0  0  0]
 [ 0  7  1  0]
 [ 0  1 13  0]
 [ 0  0  1 15]]

Classification Report:
                   precision    recall  f1-score   support

          healthy       1.00      1.00      1.00        13
  Tea leaf blight       0.88      0.88      0.88         8
Tea red leaf spot       0.87      0.93      0.90        14
     Tea red scab       1.00      0.94      0.97        16

         accuracy                           0.94        51
        macro avg       0.94      0.94      0.93        51
     weighted avg       0.94      0.94      0.94        51


Accuracy: 0.9411764705882353
```

Figure 12: confusion matrix, classification report, accuracy using RF

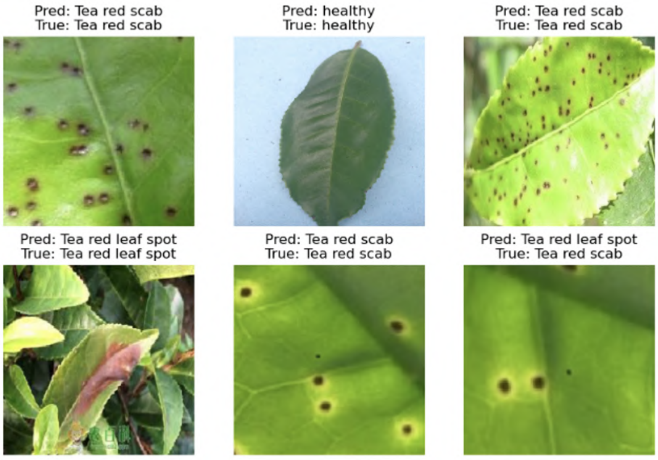


Figure 13: Prediction Result Using RF

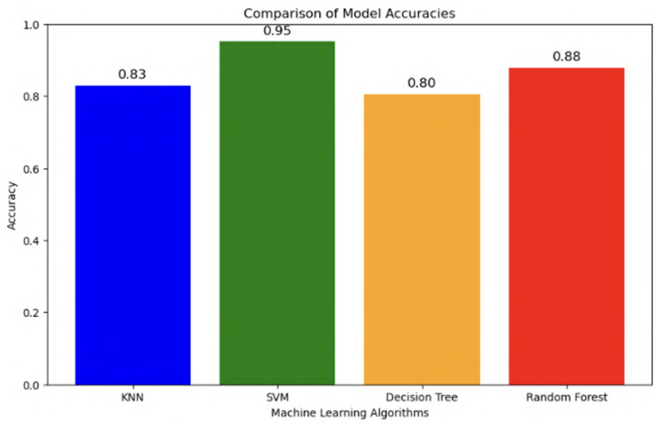


Figure 14: Accuracy Comparisons of ML algorithms

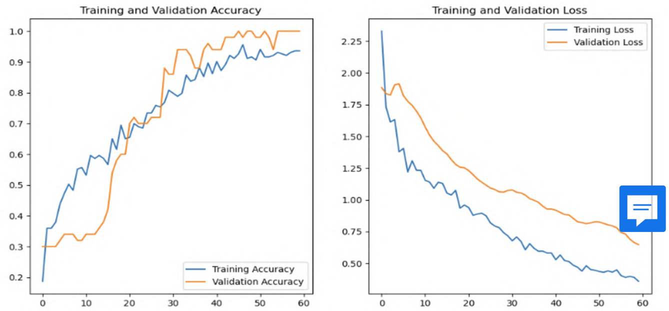


Figure 15: Accuracy and Loss of CNN for Training and Validation

```
Predictions:
 [1 3 2 1 1 0 2 3 2 1 2 1 1 3 1 3 0 0 3 0 1 3 0 3 1 3 2 2 3 3 3 0 2 1 1 0 1
 3 1 1 3 1 1 1 3 1 0 2 3 1]
Labels:
 [1 3 2 1 1 0 2 3 2 1 2 1 1 3 1 3 0 0 3 0 1 3 0 3 1 3 2 2 3 3 3 0 2 1 1 0 1
 3 1 1 3 1 1 1 3 1 0 2 3 1]
 Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         8
           1       1.00      1.00      1.00        19
           2       1.00      1.00      1.00         8
           3       1.00      1.00      1.00        15

    accuracy                           1.00        50
   macro avg       1.00      1.00      1.00        50
weighted avg       1.00      1.00      1.00        50
```

Figure 16: Classification Report of CNN

4.2. *Deep Learning Algorithms*

We evaluated the performance of several deep learning models, including CNN, VGG16, VGG19, MobileNetV2, ResNet34, AlexNet, and EfficientNetB0, for tea leaf disease classification

The VGG16 model achieve average accuracy of 87% with a weighted precision and F1-score of 0.87. Figures 28–30 present the accuracy and loss curves, the classification report, and sample prediction results obtained using the VGG16 model.

The VGG19 model achieved an accuracy of 89%, with weighted precision, recall, and F1-score of 0.90. While the

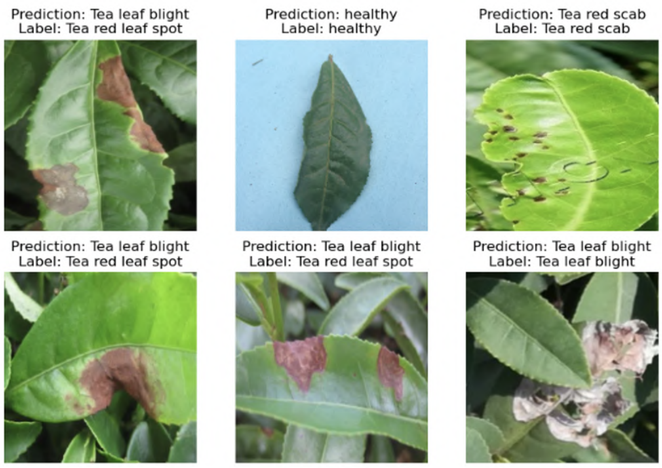


Figure 17: Prediction Result using CNN

	loss	accuracy	val_loss	val_accuracy
55	0.402770	0.926108	0.743042	1.0
56	0.388877	0.921182	0.730690	1.0
57	0.397081	0.931035	0.690546	1.0
58	0.388604	0.935961	0.663649	1.0
59	0.358417	0.935961	0.646653	1.0

Figure 18: Model Evaluation of CNN

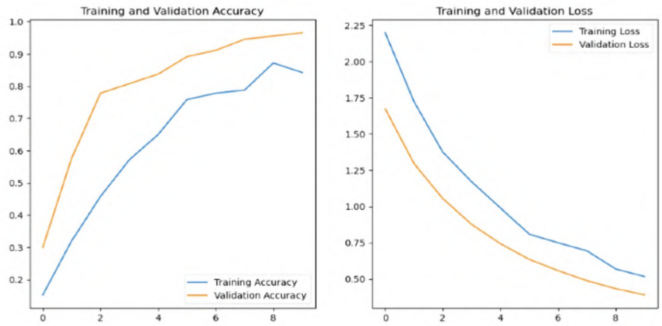


Figure 19: Accuracy and Loss of VGG16 for Training and Validation

```
1/1 [==============================] - 6s 6s/step
Predictions:
 [1 0 0 0 0 1 0 3 1 1 0 3 3 0 3 3 1 2 1 0 3 2 3 1 0 2 1 2 3 2 2 1]
Labels:
 [1 0 0 0 2 1 0 3 1 1 0 3 3 0 3 3 1 2 1 0 3 2 3 1 0 2 1 2 3 2 2 1]

              precision    recall  f1-score   support

           0       0.89      1.00      0.94         8
           1       1.00      1.00      1.00         9
           2       1.00      0.86      0.92         7
           3       1.00      1.00      1.00         8

    accuracy                           0.97        32
   macro avg       0.97      0.96      0.97        32
weighted avg       0.97      0.97      0.97        32
```

Figure 20: Classification Report of VGG16

Figure 21: Prediction Result using VGG16

```
Confusion Matrix:
 [[ 9  3  0  0]
 [ 1  6  0  0]
 [ 0  1 12  0]
 [ 0  0  0 19]]

Classification Report:
                   precision    recall  f1-score   support

   Tea leaf blight       0.90      0.75      0.82        12
Tea red leaf spot       0.60      0.86      0.71         7
      Tea red scab       1.00      0.92      0.96        13
           healthy       1.00      1.00      1.00        19

          accuracy                           0.90        51
         macro avg       0.88      0.88      0.87        51
      weighted avg       0.92      0.90      0.91        51


Accuracy: 0.9019607843137255
```

Figure 22: Classification Report of VGG19

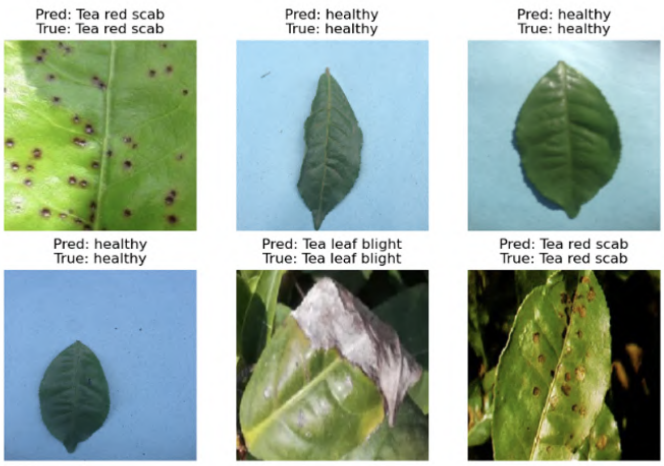


Figure 23: Prediction Result using VGG19

model showed slightly lower performance for Tea Leaf Blight and Tea Red Leaf Spot, it performed exceptionally well in classifying Healthy and Tea Crimson Scab samples, as reflected in the confusion matrix. Figures 31 and 32 present the classification report and sample predictions obtained using the VGG19 model.

MobileNetV2 achieved 90% accuracy with a weighted F1-score of 0.91. The model performed best on Healthy and Tea Crimson Scab classes, with slightly lower performance for Tea Leaf Blight and Tea Red Leaf Spot. Figures 33 and 34 show the classification report and sample predictions.

ResNet34 achieved an overall accuracy of 32%, with a weighted precision of 0.15, recall of 0.32, and F1-score of 0.17. The model struggled with Tea Leaf Blight and Tea Red Leaf Spot but performed better on Tea Crimson Scab. Training and validation accuracy curves fluctuated significantly, indicating instability, while loss curves showed reduced training loss but poor generalization. Figures 35–37 present the accuracy and loss curves, classification report, and sample predictions for ResNet34.

```
Confusion Matrix:
 [[ 9  3  0  0]
 [ 1  6  0  0]
 [ 0  1 12  0]
 [ 0  0  0 19]]
```

Classification Report:

	precision	recall	f1-score	support
Tea leaf blight	0.90	0.75	0.82	12
Tea red leaf spot	0.60	0.86	0.71	7
Tea red scab	1.00	0.92	0.96	13
healthy	1.00	1.00	1.00	19
accuracy			0.90	51
macro avg	0.88	0.88	0.87	51
weighted avg	0.92	0.90	0.91	51

Accuracy: 0.9019607843137255

Figure 24: Classification Report of MobileNetV2

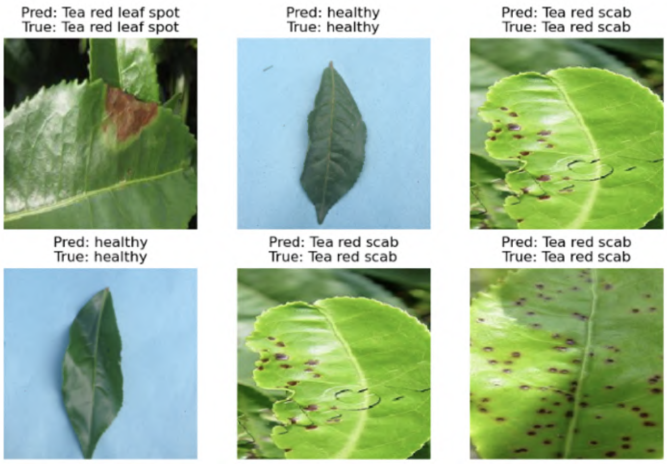


Figure 25: Prediction Result using MobileNetV2

Figure 26: Accuracy and Loss of ResNet34 for Training and Validation

Classification Report:

	precision	recall	f1-score	support
healthy	0.33	0.12	0.18	8
Tea leaf blight	0.00	0.00	0.00	19
Tea red leaf spot	0.00	0.00	0.00	8
Tea red scab	0.32	1.00	0.48	15
accuracy			0.32	50
macro avg	0.16	0.28	0.17	50
weighted avg	0.15	0.32	0.17	50

Figure 27: Classification Report of ResNet34

Figure 28: Prediction Result using ResNet34

AlexNet34 achiev 9

EfficientNetB0 achieved 96% accuracy with a weighted precision, recall, and F1-score of 0.96. The model performed exceptionally well across all classes, particularly Healthy and Tea Crimson Scab, with minimal misclassifications, as shown in the confusion matrix Figures 41 and 42.

Table 5 summarizes the comparative performance of all deep learning algorithms, while Figure 43 provides an overall comparison of their accuracies.

4.3. Explainable AI techniques (LIME And GRAD CAM)

4.3.1. Lime Technique for Machine Learning Algorithm

LIME applies slight modifications to the original image. These modified versions are referred to as “perturbations.”

Figure 29: Accuracy and Loss of AlexNet34 for Training and Validation

Table 4: Evaluation Metrics Comparisons of DL algorithms

Deep Learning Algorithms	Dataset	Accuracy	Precision	Recall	F1-Score	Support
CNN	Healthy	0.92	1.00	1.00	1.00	8
	Tea Leaf Blight	0.92	1.00	1.00	1.00	19
	Tea Red Leaf Spot	0.92	1.00	1.00	1.00	8
	Tea Red Scab	0.92	1.00	1.00	1.00	15
VGG16	Healthy	0.69	0.89	1.00	0.94	8
	Tea Leaf Blight	0.69	1.00	1.00	1.00	9
	Tea Red Leaf Spot	0.69	1.00	0.86	0.92	7
	Tea Red Scab	0.69	1.00	1.00	1.00	8
VGG19	Healthy	0.901	1.00	1.00	1.00	19
	Tea Leaf Blight	0.901	0.90	0.75	0.82	12
	Tea Red Leaf Spot	0.901	0.60	0.86	0.71	7
	Tea Red Scab	0.901	1.00	0.92	0.96	13
MobileNetV2	Healthy	0.902	1.00	1.00	1.00	19
	Tea Leaf Blight	0.902	0.90	0.75	0.82	12
	Tea Red Leaf Spot	0.902	0.60	0.86	0.71	7
	Tea Red Scab	0.902	1.00	0.92	0.96	13
ResNet34	Healthy	0.36	0.36	0.33	0.12	0.18
	Tea Leaf Blight	0.36	0.36	0.00	0.00	0.00
	Tea Red Leaf Spot	0.36	0.36	0.00	0.00	0.00
	Tea Red Scab	0.36	0.36	0.32	1.00	0.48
AlexNet34	Healthy	0.25	0.25	1.00	1.00	1.00
	Tea Leaf Blight	0.25	0.25	0.00	0.00	0.00
	Tea Red Leaf Spot	0.25	0.25	0.59	0.93	0.72
	Tea Red Scab	0.25	0.25	0.75	0.50	0.60
EfficientNetB0	Healthy	0.96	0.96	1.00	1.00	1.00
	Tea Leaf Blight	0.96	0.96	0.89	1.00	0.94
	Tea Red Leaf Spot	0.96	0.96	0.93	0.93	0.93
	Tea Red Scab	0.96	0.96	1.00	0.94	0.97

Classification Report:

	precision	recall	f1-score	support
Tea leaf blight	0.00	0.00	0.00	6
Tea red leaf spot	0.59	0.93	0.72	14
Tea red scab	0.75	0.50	0.60	6
healthy	1.00	1.00	1.00	6
accuracy			0.69	32
macro avg	0.59	0.61	0.58	32
weighted avg	0.59	0.69	0.62	32

Figure 30: Classification Report of AlexNet34

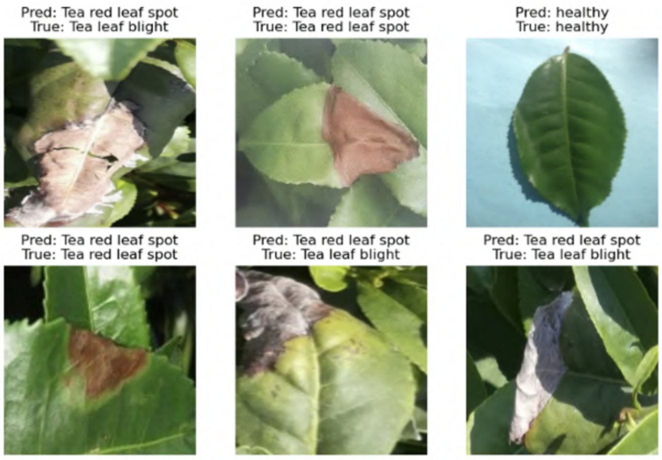


Figure 31: Prediction Result using AlexNet34

Confusion Matrix:

```
[[13  0  0  0]
 [ 0  8  0  0]
 [ 0  1 13  0]
 [ 0  0  1 15]]
```

Classification Report:

	precision	recall	f1-score	support
healthy	1.00	1.00	1.00	13
Tea leaf blight	0.89	1.00	0.94	8
Tea red leaf spot	0.93	0.93	0.93	14
Tea red scab	1.00	0.94	0.97	16
accuracy			0.96	51
macro avg	0.95	0.97	0.96	51
weighted avg	0.96	0.96	0.96	51

Accuracy: 0.9607843137254902

Figure 32: Classification Report of EfficientNetB0

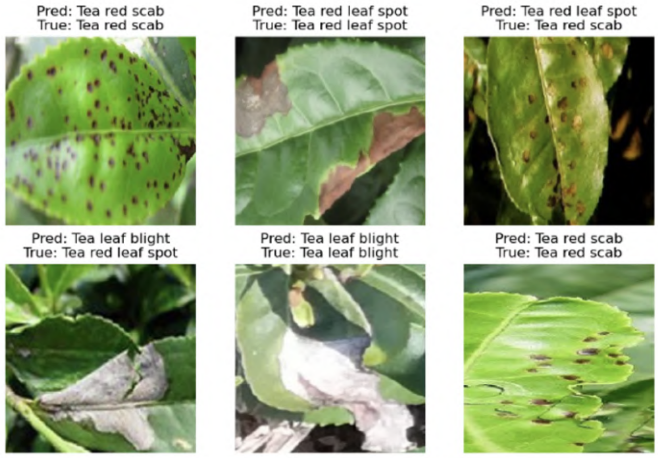


Figure 33: Prediction Result using EfficientNetB0

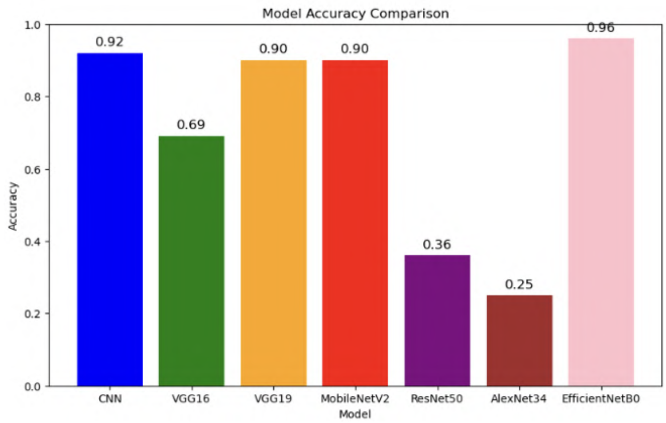


Figure 34: Accuracy Comparisons of DL algorithms

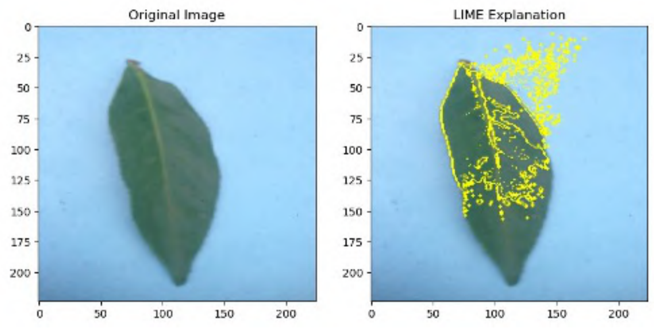


Figure 35: LIME for Disease Detection

We predict the class (the predicted class is either diseased or healthy) for each one of these perturbed images using the original model. LIME fits a simpler, more interpretable model (typically a linear model) to these perturbed images and their predictions. This local function approximates the function of the original model around the original image. MELO is a dashboard that explains the local (per-image) classifiers by learning local interpretable linear models. LIME explanation which highlights those regions in the image which contributed to the prediction using the above coefficients

For example in the Figure 44, the LIME explanation indicates the model only looked at the yellows of the leaf to predict it was diseased. This means these areas may have visual clues such as spots, discoloration or unusual texture that can be a sign of disease.

4.3.2. Grad-CAM Technique for Deep Learning Algorithm

An example of Grad-CAM visualization for a leaf disease detection model is shown in Figure 45. Grad-CAM (Gradient-weighted Class Activation Mapping) is an intuitive visual class-discriminative localization which is enabled by gradients passed through the classifier- end of the neural network found. In this case, the Grad-CAM heatmap shows that the model is concentrating on a tiny region in the middle of the leaf when deciding its class as healthy. The colors center heat indicates that this area is a powerful ruled area in this decision generated with the help of a model. This means that the model is likely identifying individual visual features like color, texture, or shape in this section because it needs those features to classify it.

4.4. Web Interface Design

We developed an internet-primarily based app using Streamlite for predicting tea leaf disease. The app affords an intuitive interface, comprising 3 pages: domestic, approximately, and Prediction. The home page includes a sidebar for navigation. The about web page presents information about the dataset, the assignment's content, the group, and goals. The Prediction page allows customers to upload images of tea leaves from their local tool, which may be labeled as wholesome, Red leaf spot, tea red scab, or tea leaf blight. After uploading, a Button shows the image alongside the anticipated sickness. This device makes use of EfficientNetB0, the most efficient model evaluated, to provide accurate disorder prediction.

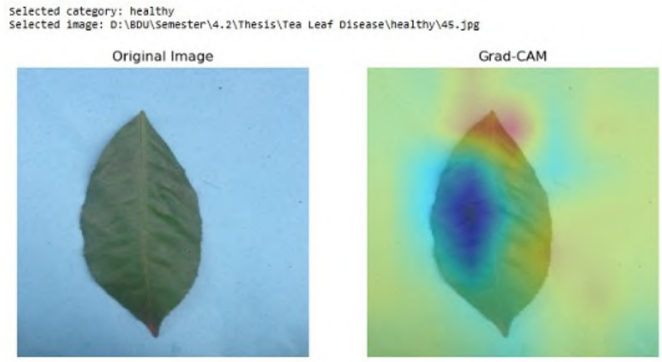


Figure 36: Grad-Cam for Disease Detection

4.4.1. Homepage

A very important sections of the application which defines what this application deals with is the homepage. It contains the title, "Tea Leaf Disease Detection System" that summarizes the purpose of the user platform, as well as the relevance, along with a cool and colorful image.
It is very well captured in their mission statement: Users can upload pictures of tea leaves to receive accurate analysis and actionable recommendations. It helps to detect and manage diseases of plant at the early stage resulting which increases the production and health of the crop.
A simplified step by step guide explains how the system works, where the users upload a picture(s), the system analyzes users crops for diseases, and returns the recommendations with its outcomes. The clarity makes use easy for users of any kind.
The first thing we see on the homepage (Figure 46) is the system's core strength at a glance (eg. high accuracy, simplicity and real-time feedback) to persuade the user to trust and use the platform.

4.4.2. Dashboard

The "Dashboard" page (Figure 47) is the main landing page of "Tea Leaf ailment Detection device" which has a drop down menu provided for easy navigation. After that, users can then choose one of 3 options so that you can gain access to the main feature, Tea Leaf sickness Detection, about for platform information, and domestic for the system review. The layout is easy, dark-themed, and accompanied by a pink-highlighted dropdown menu, which ensures a clean studying and use experience. There is a near button in the pinnacle-proper, so people can quickly dismiss the interface: making it powerful and smooth to apply.

4.4.3. About Page

A longer description of the "Tea Leaf Disease Detection System" and its main components can be located on the About Page (Figure 48).

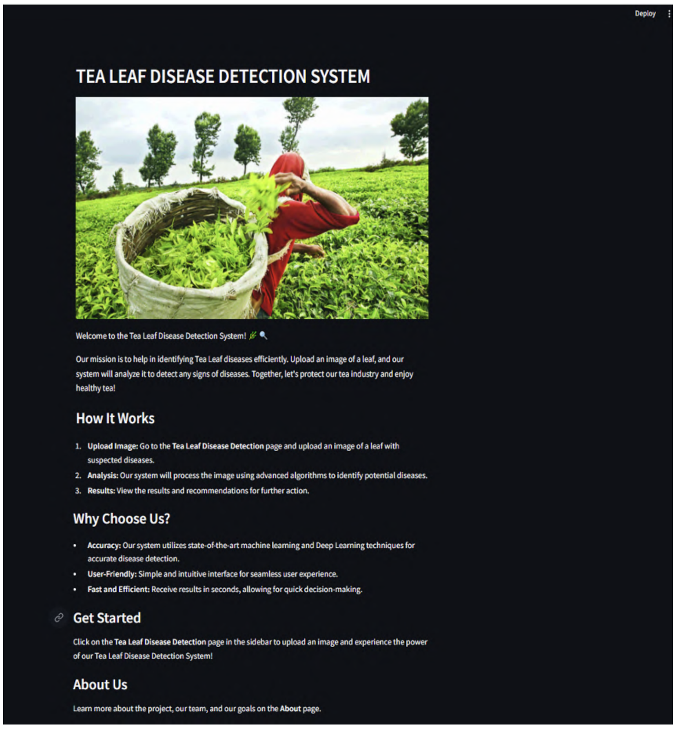


Figure 37: Homepage

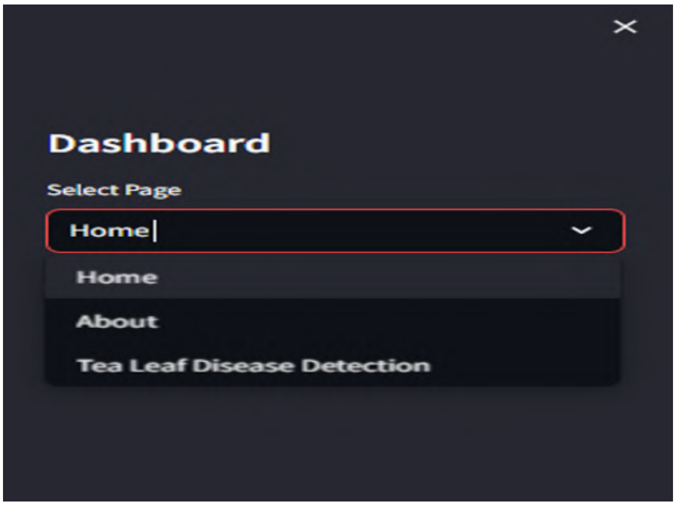


Figure 38: Dashboard

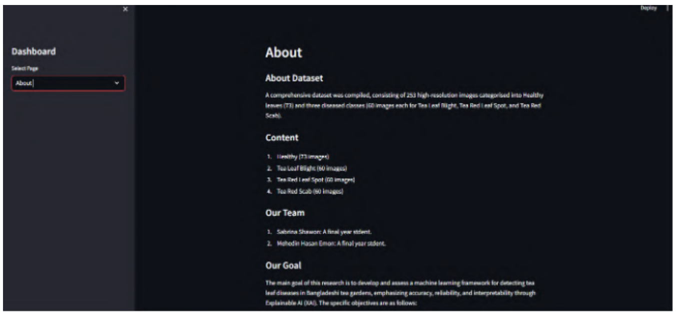


Figure 39: About Page

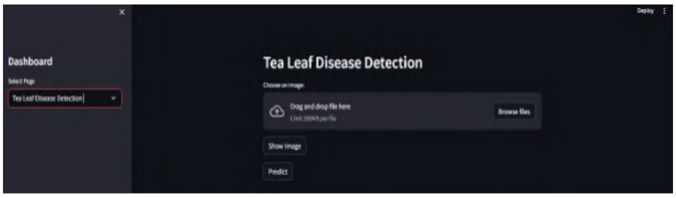


Figure 40: Prediction Page

About Dataset. The collection contains high-resolution images from four classes, including Tea Red Leaf Spot, Tea Leaf Blight, Healthy, and Tea Red Scab.

Content. Content Lists the categories in the dataset, namely: Tea Red Leaf Spot, Tea Leaf Blight, Healthy, and Tea Red Scab.

Our Team. Highlights the main contributors.

Goal. Describes the objectives of this project, which are:

- Develop the Early Detection Machine Learning Framework for Tea Leaves illness.
- Disease Detection precisely and provide practical advice related to disease to the growers accurately.
- Influence towards different AI applications in Bangladesh agricultural industries.

4.4.4. *Prediction Page*

The working procedure of the prediction page interface is given in the Figure 49, 50, and 51.

- The user interacts with the Prediction Page and clicks the ”Browse Files” button to upload an image. (Fig.49)
- A file explorer dialog opens, allowing the user to navigate to the desired directory and select an image file (e.g., rs7.jpg). (Fig.50)
- The uploaded tea leaf image is displayed in the application, confirming successful upload and readiness for prediction. (Fig.51)

The predicted result based on the image selected (leaf condition) is given below in Table 6.

5. IMPACT ON SOCIETY, ENVIRONMENT AND SUSTAINABILITY

5.1. *Impact on Society*

There are giant social ramifications while tea leaf ailments are detected using Convolutional Neural Networks (CNN), device studying (ML), and Explainable artificial Intelligence (XAI). Through precise and active detection of illnesses inclusive of Tea Leaf Blight, Tea red Leaf Spot, and Tea crimson Scab, those technologies allow tea manufacturers to take preventative measures, which decrease crop losses and boost

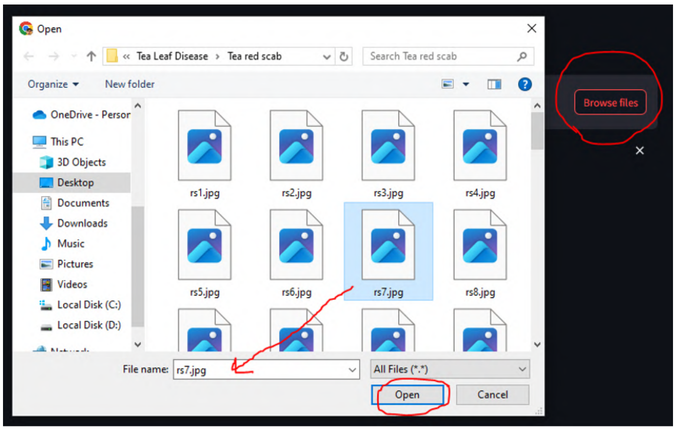


Figure 41: Browse Image (File Selection Step)

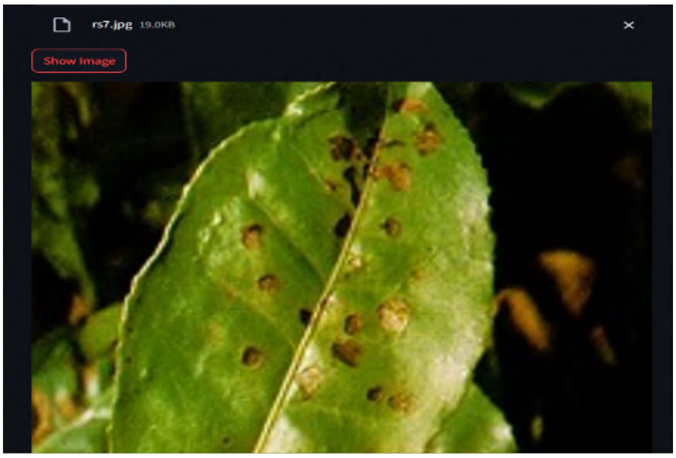


Figure 42: Display Image

agricultural output. Farmers take advantage of this, in particular small-scale farmers who are extra susceptible to the consequences of crop diseases. The initiative additionally encourages using this method in the agriculture sector with the aid of promoting a information-pushed agricultural strategy. that is in line with the wider social objective of assisting sustainable farming strategies and making certain meals safe.

5.2. *Impact on Environment*

The use of CNN, ML, and XAI in tea leaf disease detection has a comparatively lower environmental footprint than traditional methods. Digital tools reduce the need for extensive physical inspections and manual analysis, thus minimizing resource consumption. However, running complex models like CNNs involves significant computational power, which could increase energy usage. To offset this, adopting energy-efficient computational practices, such as optimized model architectures and sustainable data storage solutions, can help mitigate environmental impacts. This balance ensures that technological advancements contribute to environmental conservation.

5.3. *Ethical Aspects*

Ethical considerations are paramount when implementing CNN and ML models in agriculture. XAI approaches shall help to bring in transparency to forecasts, ensuring that farmers and agronomists believe the suggestions by the system. Efforts must also be made to avoid biases in the datasets or algorithms that could result in unfair treatment of specific regions or farmers. This project emphasizes ethical integrity by maintaining data security and transparency, ensuring fair and equitable outcomes.

5.4. *Sustainability*

To assure the sustainability of CNN, ML, and XAI systems in agricultural applications, the development of adaptable and efficient frameworks is crucial. These models should be capable of handling diverse datasets and evolving alongside advancements in technology. Regular updates to the models and energy-efficient processing techniques are essential for maintaining relevance while minimizing resource usage. Integrating these technologies into existing agricultural workflows ensures their practicality and long-term viability. By fostering collaborations with agricultural experts, this project creates a sustainable pathway for using AI-driven solutions to improve farming practices.

6. CONCLUSION AND FUTURE WORK

6.1. *Conclusion*

The paper explores the use of CNN, ML and Logical AI (XAI) approaches to distinguish tea leaf diseases, including Tea Leaf Curse, Tea Ruddy Leaf Spot and Tea Ruddy Scab. Deep learning adapts existing machine learning techniques to create a model that demonstrates superior performance in classifying tea leaf conditions. This is achieved through explainability with the LIME and GRAD CAM visualizations, which provide two different views of how different perspectives have contributed to the model's expectations. These explanations help bridge the gap between cutting-edge innovation and viable agribusiness applications, allowing growers to adopt and have confidence in the framework's results. The main contributions of this work are:

- **Model Development and Performance:** The CNN and ML models were designed, validated, and tested for a number of datasets to enable the diagnosis of diseases with high accuracy and robustness.

- **Explainable AI:** Combination of LIME and GRAD CAM visualizations made the predictions interpretable, fostering trust among users by explaining how features like texture, color, and shape contribute to disease identification.

- **Feature Importance Analysis:** The study identified critical features contributing to the classification, offering valuable insights for future research and practical disease management strategies.

6.2. Future Work

The diseases analysed were Tea Leaf Blight, Blue mould of tea leaves, Tea Red Leaf Spot and Tea Red Scab. Even if, in general, the DL(deep-learning) and ML(machine-learning) models used (particularly CNNs) have yield promising results, there are still quite a few limitations and room for improvements.

A key disadvantage is the absence of a mobile app. Next, a mobile app is in the works since the current study is centered on web-based prediction. In this mobile software integrated to train these models, farmers will have the opportunity to take and check leaf photos through their mobile phones according to the species of the plants. The system will provide actionable statrep data, such as the type, severity and recommended treatments of the illness, to ensure practical applicability in agricultural settings.

The dataset is also limited in that the samples have little variability in terms of environmental condition, lighting, and tea leaf variation, giving potential to improve performance if the dataset is extended to include samples with such characteristics. Improving the robustness and ability of model to adapt better in real-life scenarios.

In future work, the user feedback could be integrated and explainable AI methods could be used to further enhance prediction transparency. This should inspire trust between farmers and agricultural experts, and appeal to AI-driven solutions.

This project aims to overcome these limitations by providing tea growers with an easy-to-use, low-cost, and scalable tool to support sustainable agricultural practices and reduce crop loss.

Acknowledgements

Thanks to ...

Appendix A. Appendix title 1

Appendix B. Appendix title 2

References

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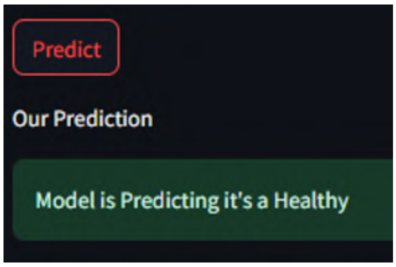
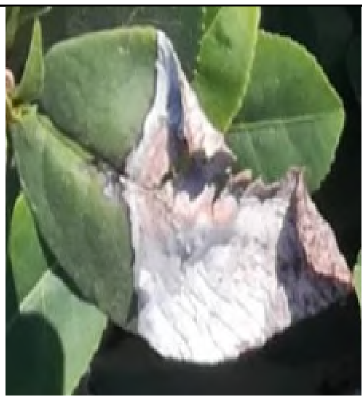
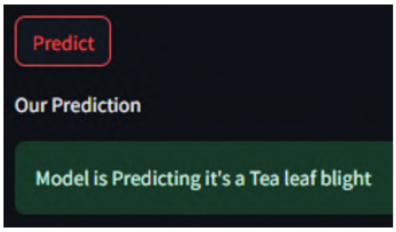
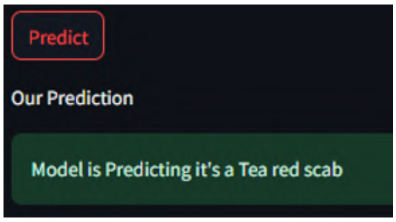
Table 5: Predicted Result

Leaf Condition	Image	Prediction
Healthy		
Tea Leaf Blight		
Tea Red Scab		
Tea Red Leaf Spot